

Data Cleaning

In this step, the dataset was cleaned to handle missing values and remove unnecessary columns.

The following actions were performed:

- Missing values in the Age column were filled using the mean value
- The Cabin column was dropped as it contained too many missing values

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
#load dataset
df = pd.read_csv("/content/Titanic-Dataset.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath	female	35.0	1	0	113803	53.1000	C123	S

```
#Filling missing age values with mean
df["Age"].fillna(df["Age"].mean())
```

	Age
0	22.000000
1	38.000000
2	26.000000
3	35.000000
4	35.000000
...	...
886	27.000000
887	19.000000
888	29.699118
889	26.000000
890	32.000000
891 rows × 1 columns	
dtype: float64	

```
#Dropping irrelevant groups
df.drop(columns=["Cabin"])
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	AgeGroup
0	1	0	3	Braund, Mr. Owen Harris	male	22.000000	1	0	A/5 21171	7.2500	S	YoungAdult
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.000000	1	0	PC 17599	71.2833	C	Adult
2	3	1	3	Heikkinen, Miss. Laina	female	26.000000	0	0	STON/O2. 3101282	7.9250	S	YoungAdult
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.000000	1	0	113803	53.1000	S	Adult
4	5	0	3	Allen, Mr. William Henry	male	35.000000	0	0	373450	8.0500	S	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.000000	0	0	211536	13.0000	S	YoungAdult
...	...	...	...	...	...	...	...	...	...	...	...	...

Analysis

In this step, different factors affecting survival were analyzed.

1. Survival Rate by Age Group

- Age was divided into groups such as Child, Teen, Adult, etc.
- Children showed higher survival rates compared to older passengers

```
bins = [0, 12, 20, 40, 60, 100]
labels = ['Child', 'Teen', 'Adult', 'Mid-age', 'Senior']

df['AgeGroup'] = pd.cut(df['Age'], bins=bins, labels=labels)

df.groupby('AgeGroup', observed=False)['Survived'].mean()
```

Survived	
AgeGroup	
Child	0.579710
Teen	0.381818
Adult	0.397403
Mid-age	0.390625
Senior	0.227273

dtype: float64

2. Survival Rate by Embarkation Port

- Passengers were grouped based on embarkation ports (C, Q, S)
- Survival rates varied depending on boarding location

```
df.groupby('Embarked')['Survived'].mean()
```

Survived	
Embarked	
C	0.553571
Q	0.389610
S	0.336957

dtype: float64

3. Survival Rate by Family Size

- Family size was calculated using SibSp and Parch
- Passengers with smaller families had better survival chances

```
df['FamilySize'] = df['SibSp'] + df['Parch']
df.groupby('FamilySize')['Survived'].mean()
```

Survived	
FamilySize	
0	0.303538
1	0.552795
2	0.578431
3	0.724138
4	0.200000
5	0.136364
6	0.333333
7	0.000000
10	0.000000

dtype: float64

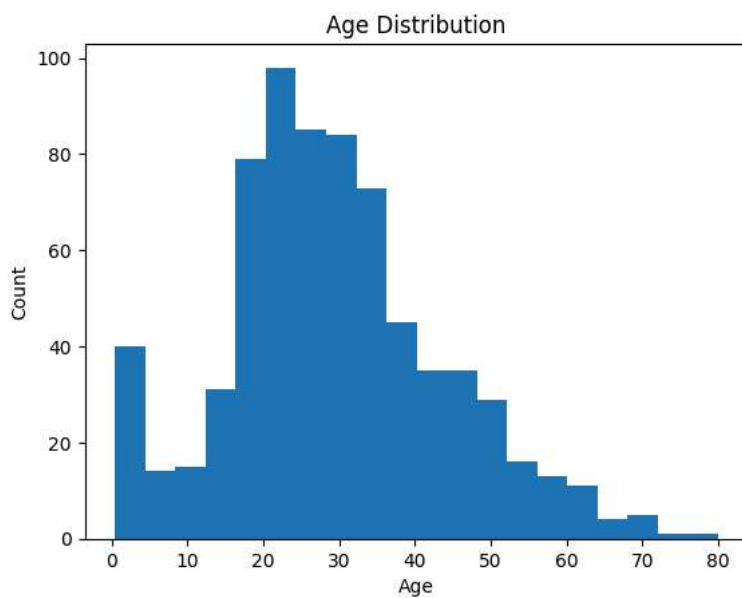
Data Visualization

Visualization helps in understanding patterns more clearly.

1. Age Distribution (Histogram)

- Shows distribution of passengers by age
- Most passengers were in the 20–40 age range

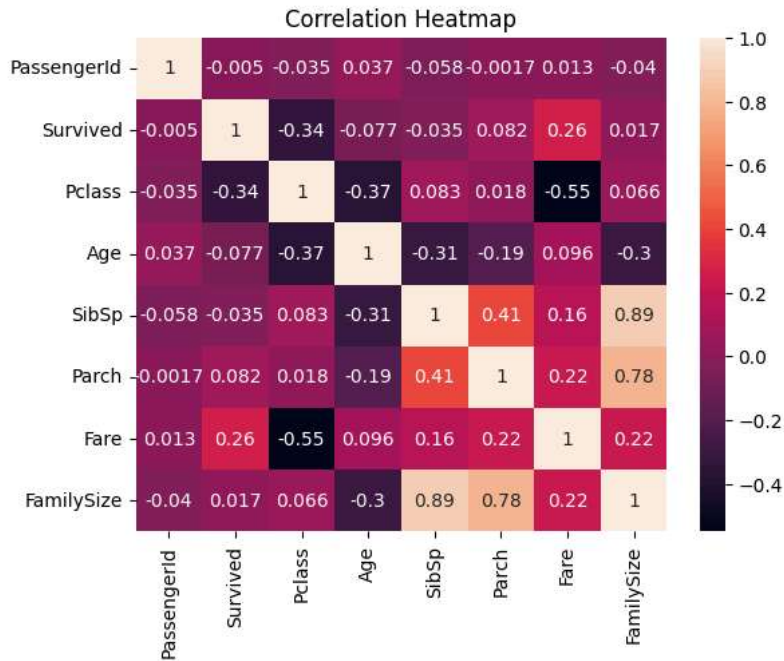
```
plt.hist(df['Age'], bins=20)
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



2. Heatmap of Correlations

- Displays relationships between numerical features
- Passenger class and fare influenced survival
- Age had a weaker relationship

```
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True)
plt.title('Correlation Heatmap')
plt.show()
```



3. Survival by Family Size (Bar Plot)

- Shows survival rate across different family sizes
- Smaller families had higher survival rates

```
sns.barplot(x='FamilySize', y='Survived', data=df)
plt.title('Survival Rate by Family Size')
plt.xlabel('Family Size')
plt.ylabel('Survival Rate')
```

```
Text(0, 0.5, 'Survival Rate')
```

